

PHYS 357: Honours Quantum Physics 1

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Abstract

NOT FINISHED

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1 Introduction

Textbook used is JS Townsend A Modern Approach to Quantum Mechanics, 2nd ed., University Science Books 2012. Taught by Bill Coish. Mondays, Wednesdays and Fridays from 9:35 to 10:25 in Steward Biology Building N2/2. Course grading is divided as follows:

- 10% problem sets (must be handwritten scans).
- 40% quizzes (on wednesdays, best 8 of 12).
- 20% midterms (best of 2, october 2nd and november 6th).
- 30% final exam.

2 Stern-Gerlach Experiments

2.1 Angular Momentum and Magnetic Moments

Definition 1 (Orbital Angular Momentum). *The orbital angular momentum of a particle is*

$$\vec{L} = \vec{r} \times \vec{p}, \quad (1)$$

where $\vec{r}$ is the position vector and $\vec{p} = m\vec{v}$ is the linear momentum.

For motion confined to the x - y plane, only the z component survives. On a circular path of radius r with speed v ,

$$L_z = mvr. \quad (2)$$

For a rigid body rotating about a fixed axis this becomes $L = I_0\omega$, with I_0 the moment of inertia and ω the angular velocity.

Magnetic moment of a circulating charge

Consider a point particle of mass m and charge q in a circular orbit of radius r and period T . Three steps connect this to a magnetic moment.

1. The charge passes any point on the orbit once per period, so it is equivalent to a steady current

$$I = \frac{q}{T} = \frac{qv}{2\pi r}. \quad (3)$$

2. A current loop of area $A = \pi r^2$ has magnetic moment

$$\mu = \frac{IA}{c}. \quad (4)$$

3. Substituting gives

$$\mu = \frac{1}{c} \left(\frac{qv}{2\pi r} \right) (\pi r^2) = \frac{q}{2mc} mvr = \frac{q}{2mc} L_z. \quad (5)$$

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Result 1 (Gyromagnetic Ratio). *The magnetic moment is proportional to the angular momentum,*

$$\vec{\mu} = \gamma \vec{L}, \quad \gamma = \frac{q}{2mc}. \quad (6)$$

The constant γ is the gyromagnetic ratio.

Remark 1. *This is a purely classical result. It matters here because measuring $\vec{\mu}$ is how we measure $\vec{L}$, and it is what the Stern-Gerlach apparatus exploits.*

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A Appendix

B Useful Links